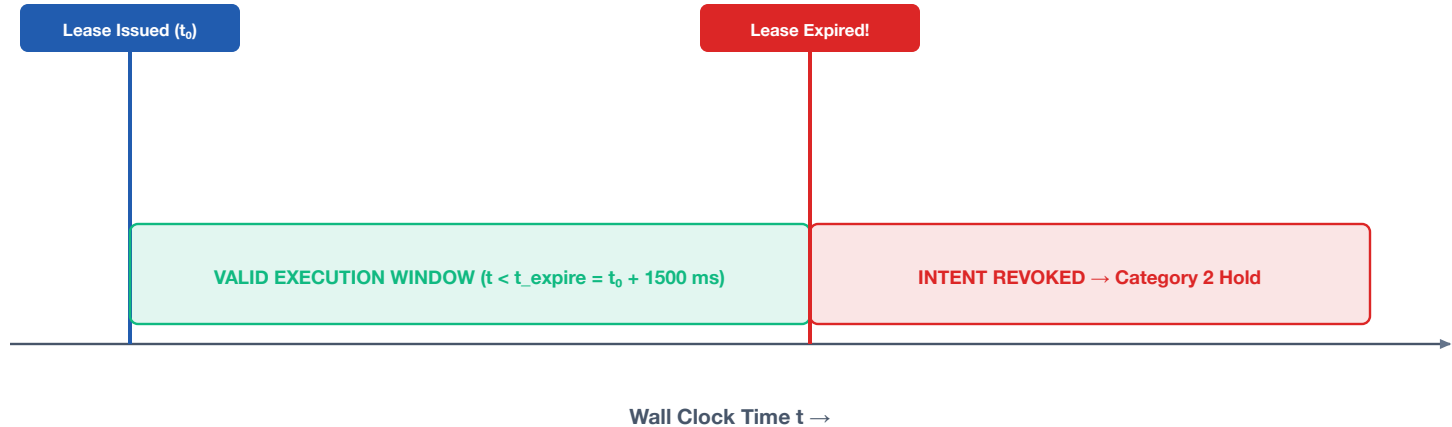


EXPIRING INTENT LEASE LIFECYCLE & DEAD-MAN SWITCH

Preventing Zombie Goal Execution: Every High-Level Intent Has a Finite Hardware Validity Window



THE INTENT LEASE CONTRACT

1. High-level reasoning models must never issue permanent goals; all intents carry an explicit monotonic timestamp deadline.
2. If the VLM crashes or hangs, the real-time MCU safety layer detects lease expiry and brings the robot to a controlled stop.
3. The agent never moves on stale intent without an active renewal heartbeat.